

set 1:

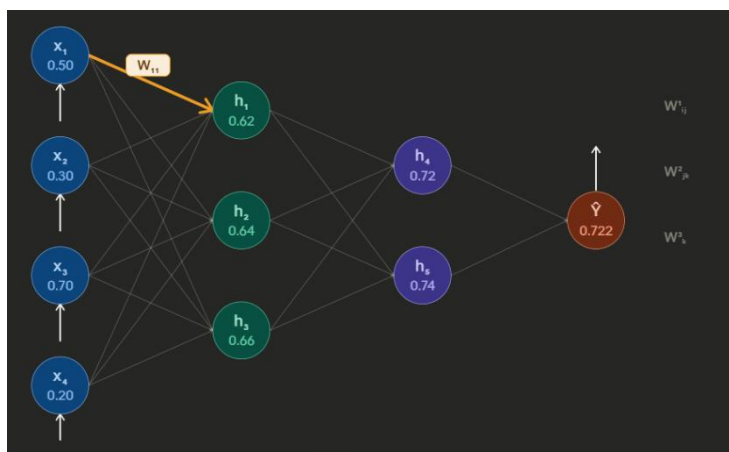
short question (2 marks):

1. What is vanishing gradient problem?
2. What is tensor in linear algebra? Give example.
3. What is Dimension Reduction? Write the name of two methods of Dimension Reduction.
4. What happens if the learning rate is too high or too low? Explain with figure.
5. What is data normalization? Write the name of different techniques to achieve Data normalization.
6. What is activation function? Give some examples of activation functions.
7. Write some application of Deep learning.

short question (2.5 marks):

1. What is the differences between ANN, CNN & RNN
2. What is the differences between Dense and Sparse features in dataset?
3. What is the differences between Batch Gradient Descent and Stochastic Batch Gradient Descent?
4. What is Bias and Variance? What is the scenario of Bias and Variance on Model Overfitting and Underfitting?

short question (3 marks):



1.Explain the figure properly.

Assume $Y \neq Y'$. Write the formula for update W_{11} by using chain rule equation. How will you get $Y = Y'$? explain your answer.

1. How does CNN works? Explain with proper diagram.
2. Describe the architecture of LSTM with proper diagram.
3. What is perceptron? Write the perceptron learning algorithm. Explain the geometric view of perceptron.

set 2

short question (2 marks):

1. What are the main applications of deep learning?
2. What is data normalization? Write the name of different techniques to achieve Data normalization.
3. How to remove Vanishing Gradient Problem?
4. Write three differences between ANN, CNN & RNN.
5. WHAT ARE GLOBAL MINIMA AND LOCAL MINIMA?
6. WHEN WE CALCULATE MEAN SQUARED ERROR(MSE) AND BINARY CROSS ENTROPY? WRITE THE EQUATIONS OF THESE LOSS FUNCTION.
7. Write the equation for ReLU activation function. Give two advantages of ReLU activation function.

short question (2.5 marks):

1. Describe PCA(Principle Component Analysis) with an example.
2. What is Saturating Function? Explain it by two activation functions.
3. Explain Adagrad Optimizer.
4. Assume you have a neuron with two inputs $X = [3,7]$ that uses both the ReLU and Sigmoid activation functions separately. The neuron has the following parameters:

- Weights = [0.4, 0.6]
- Bias = 0.2

Calculate the output of the neuron during the feedforward process.

question (5 marks):

1. How does CNN work? Explain with proper diagram.
2. Describe the architecture of LSTM with proper diagram.
3. What is perceptron? Write the perceptron learning algorithm. Explain the geometric view of perceptron.
- 4.

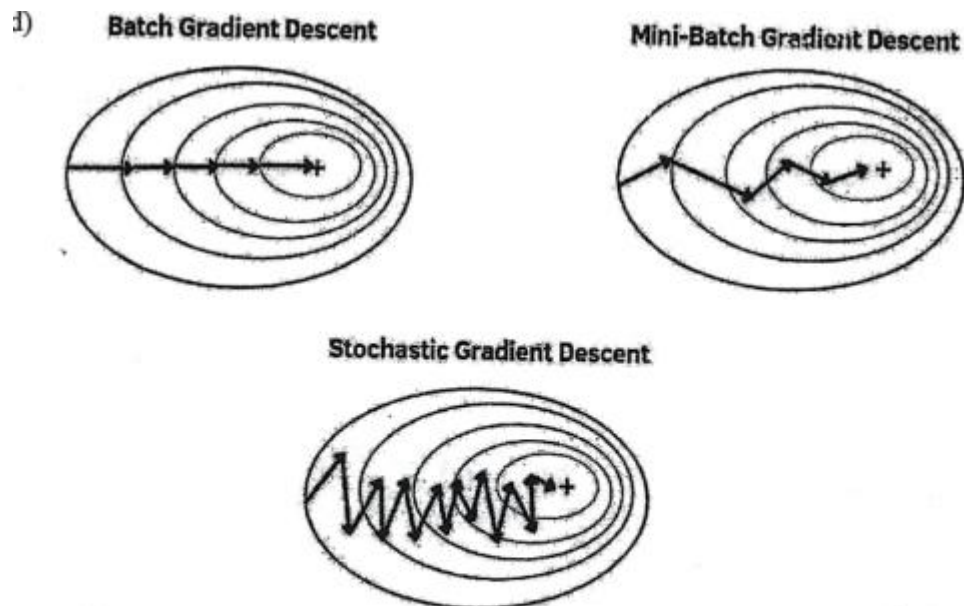


Figure: Optimizers

What is Noise in optimizer? How do you reduce the Noise? Explain with mathematical formula.